

Alpilla Enerji Üretim A.Ş.

Alpilla 600 MW Combined Cycle Power Plant

Grid Connection Data

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Status	Issued for tender / contract
Confidentiality	Restricted - Contract ALP-EPC-001 parties only

	Prepared	Checked	Approved
Function	Owner's Engineer (discipline lead)	Owner's Engineer (project engineer)	Owner's Project Director
Signature / date	signed 2026-09-11	signed 2026-09-11	signed 2026-09-11

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CASE STUDY - all organisations, data and signatures are fictional.

1. Connection

- Connection to the 380 kV TSO substation "Kıyı" (fictional name) by a 7 km double-circuit overhead line built by the Owner.
- Connection agreement signed 2026-06-20; connection capacity 640 MW export, 40 MW import.
- Line completion and energisation readiness planned by the Owner: **2028-09-30**.

2. System data at the plant 380 kV busbar

Parameter	Value
Nominal / highest system voltage	380 kV / 420 kV
Normal voltage range	360–410 kV; 342–420 kV for up to 30 min
Frequency	50 Hz; continuous 49.0–51.0 Hz; 47.5–52.5 Hz time-limited per Grid Code
Three-phase short-circuit level (2029)	maximum 40 kA, minimum 12 kA
Future maximum (2035 planning)	50 kA
X/R ratio	14 (maximum case)
Earthing	solidly earthed
Fault clearing time (main protection)	≤ 100 ms
Auto-reclose	single-phase, 1 s dead time on the connection line

3. TSO requirements (summary)

- Fault ride through, reactive capability 0.85 lagging / 0.95 leading at the connection point, primary frequency control (droop 4 %), secondary frequency control (AGC) and fast frequency response.
- Main 1 / Main 2 line differential protection from different manufacturers; busbar protection with breaker failure.
- Revenue metering class 0.2S main and check; RTU with IEC 60870-5-104 to the national control centre.
- Power quality: harmonic emission limits per the TSO's connection conditions.
- Compliance tests witnessed by the TSO before commercial operation.